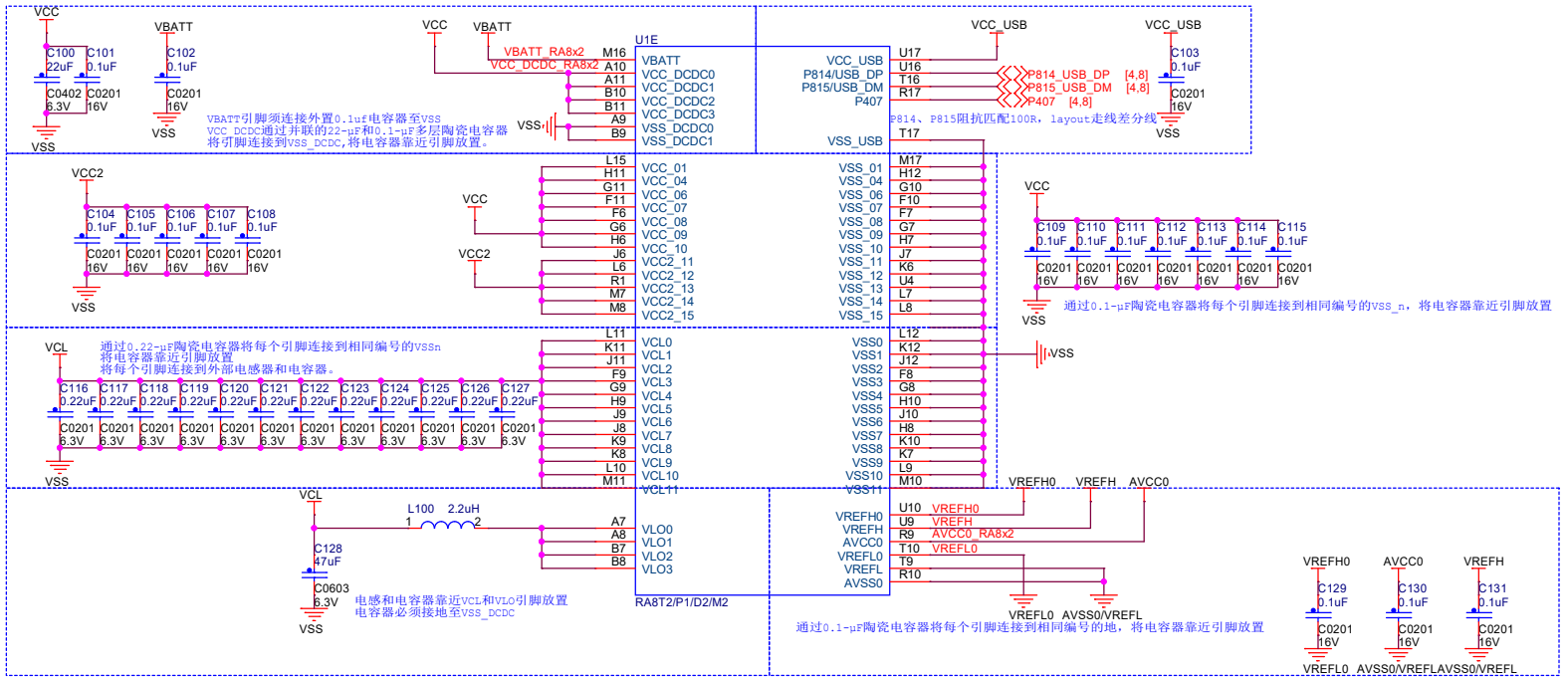
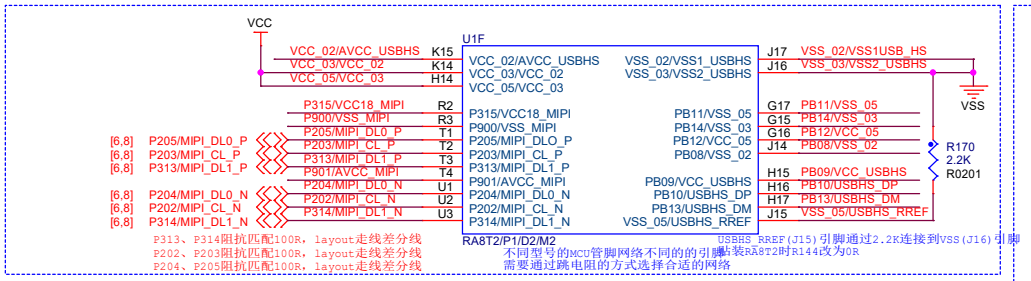


Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

核心板缺省功能阻容设置

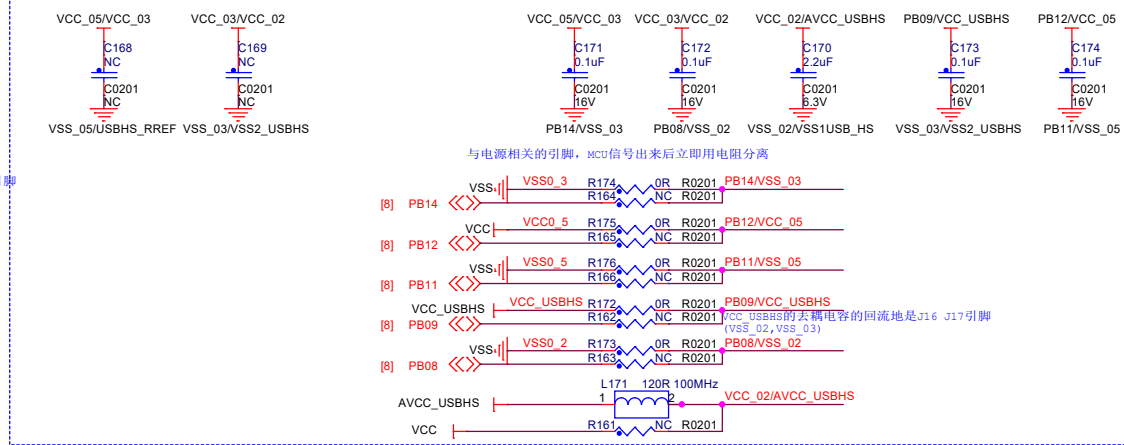
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1UF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHZ
R172-R180	NC	0R
R191-R193	NC	0R



由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2

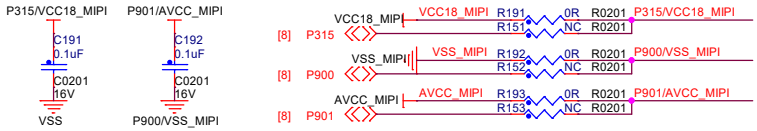
电阻选装

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



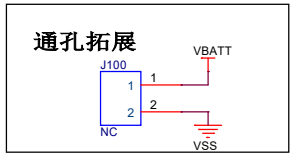
MIPI部分

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



差分线走到J1(BTB)附近再用电阻分离。

通孔拓展



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Title
野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用

Size	Document Number	Rev
A3	MCU 电源	V0.1

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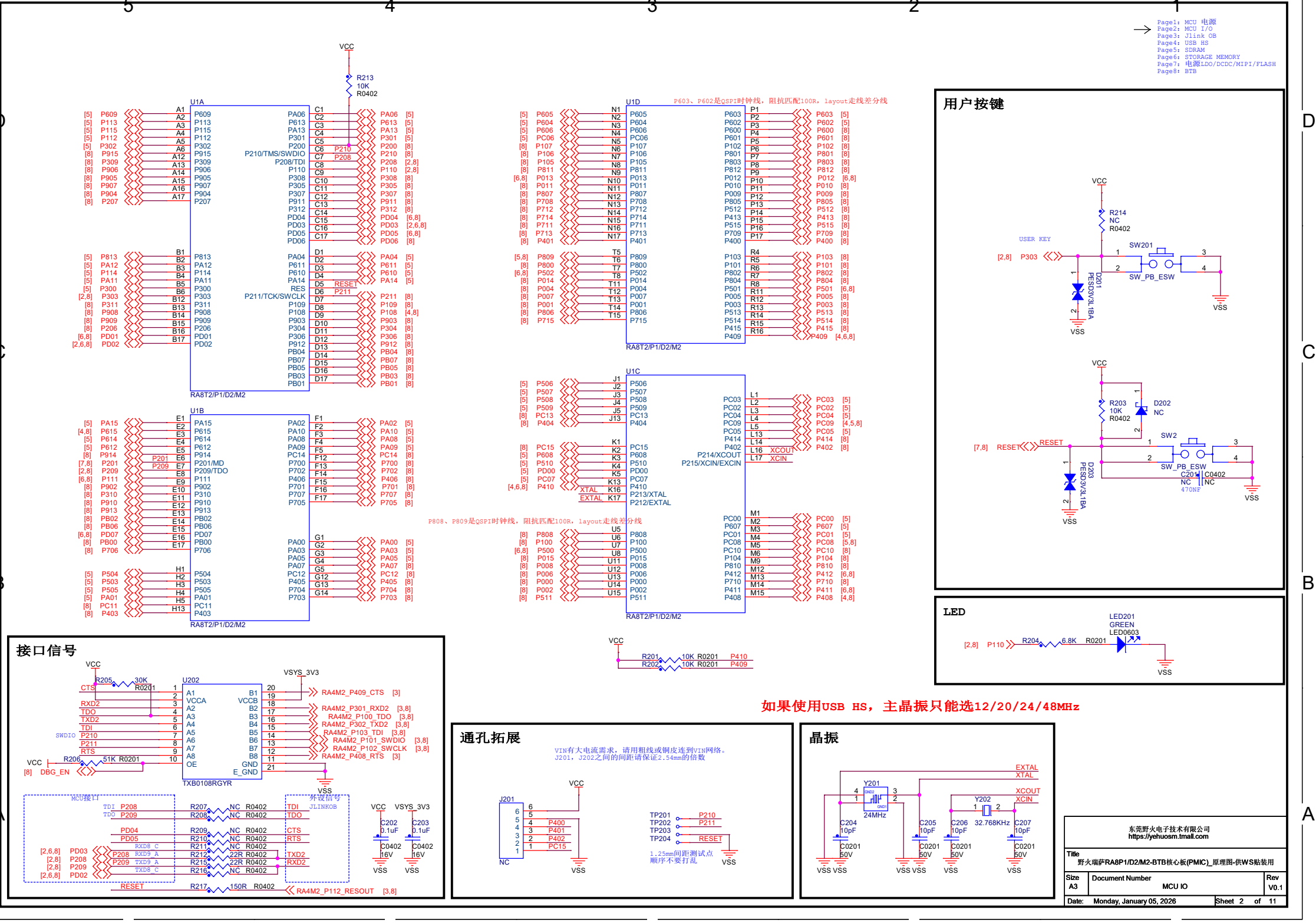
Page1: MCU 电源
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Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB
U1A
U1B
U1C
U1D
RA8T2/P1/D2/M2
P603、P602是QSPI时钟线，阻抗匹配100R，layout走线差分线
P808、P809是QSPI时钟线，阻抗匹配100R，layout走线差分线
用户按键
USER KEY
SW201
SW_PB_ESW
D201
PESD3V3L1BA
R214
NC
R0402
R203
10K
D202
SW2
C201
C0402
470NF
D203
LED
LED201
GREEN
LED0603
R204
6.8K
R0201
接口信号
U202
TXB0108RGYR
R205
30K
R206
51K
DBG_EN
MCU接口
外设信号
JLINKOB
通孔拓展
VIN有大电流需求，请用粗线或铜皮连到VIN网络。
J201，J202之间的间距请保证2.54mm的倍数
J201
1.25mm间距测试点
顺序不要打乱
晶振
Y201
24MHz
Y202
32.768KHz
C204
10pF
C205
10pF
C206
10pF
C207
10pF
C0201
50V
如果使用USB HS，主晶振只能选12/20/24/48MHz
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Title
野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用
Size
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V0.1
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Jlink OB

注意：R7FA4M2只能12MHz晶振

Page1: MCU 电源
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Page3: Jlink OB
Page4: USB HS
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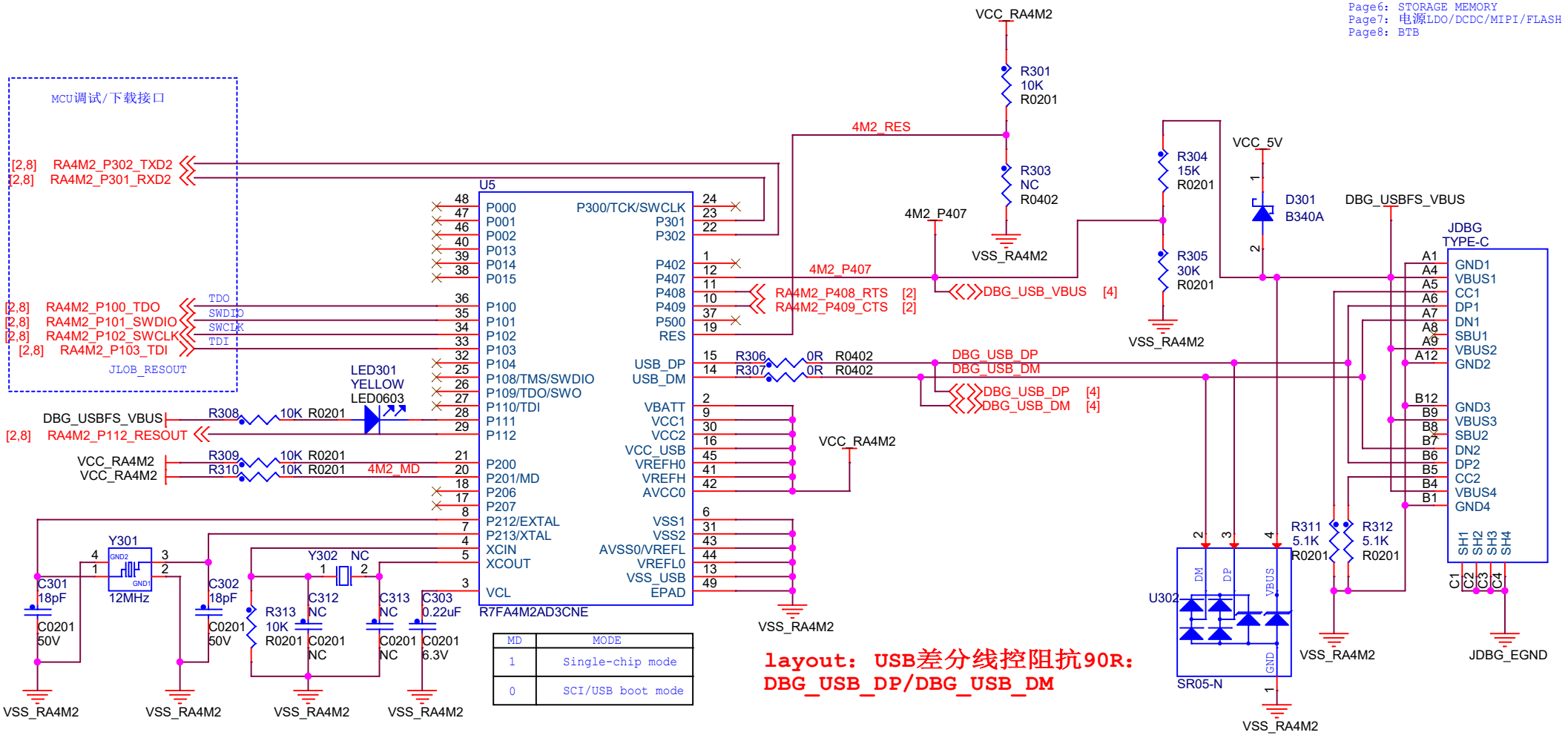
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB


MCU调试/下载接口
[2,8] RA4M2_P302_TXD2
[2,8] RA4M2_P301_RXD2
[2,8] RA4M2_P100_TDO
[2,8] RA4M2_P101_SWDIO
[2,8] RA4M2_P102_SWCLK
[2,8] RA4M2_P103_TDI
JLOB_RESOUT
TDO
SWDIO
SWCLK
TDI
U5
R7FA4M2AD3CNE
P000
P001
P002
P013
P014
P015
P100
P101
P102
P103
P104
P108/TMS/SWDIO
P109/TDO/SWO
P110/TDI
P111
P112
P200
P201/MD
P206
P207
P212/EXTAL
P213/XTAL
XCIN
XCOUT
VCL
P300/TCK/SWCLK
P301
P302
P402
P407
P408
P409
P500
RES
USB_DP
USB_DM
VBATT
VCC1
VCC2
VCC_USB
VREFH0
VREFH
AVCC0
VSS1
VSS2
AVSS0/VREFL
VREFL0
VSS_USB
EPAD
VCC_RA4M2
R301
10K
R0201
R303
NC
R0402
VSS_RA4M2
4M2_RES
4M2_P407
RA4M2_P408_RTS [2]
RA4M2_P409_CTS [2]
DBG_USB_VBUS [4]
R306 0R R0402
R307 0R R0402
DBG_USB_DP
DBG_USB_DM
DBG_USB_DP [4]
DBG_USB_DM [4]
R304
15K
R0201
R305
30K
R0201
VCC_5V
D301
B340A
DBG_USBFS_VBUS
JDBG
TYPE-C
GND1
VBUS1
CC1
DP1
DN1
SBU1
VBUS2
GND2
GND3
VBUS3
SBU2
DN2
DP2
CC2
VBUS4
GND4
SH1
SH2
SH3
SH4
C1
C2
C3
C4
R311
5.1K
R0201
R312
5.1K
R0201
U302
SR05-N
DM
DP
VBUS
GND
JDBG_EGND
LED301
YELLOW
LED0603
DBG_USBFS_VBUS
R308
10K R0201
[2,8] RA4M2_P112_RESOUT
VCC_RA4M2
VCC_RA4M2
R309
10K R0201
R310
10K R0201
4M2_MD
Y301
12MHz
C301
18pF
C0201
50V
C302
18pF
C0201
50V
Y302 NC
R313
10K
R0201
C312
NC
C0201
NC
C313
NC
C0201
NC
C303
0.22uF
C0201
6.3V
VSS_RA4M2
VSS_RA4M2
VSS_RA4M2

USB HS

如果使用USB HS，主晶振只能选12/20/24/48MHz

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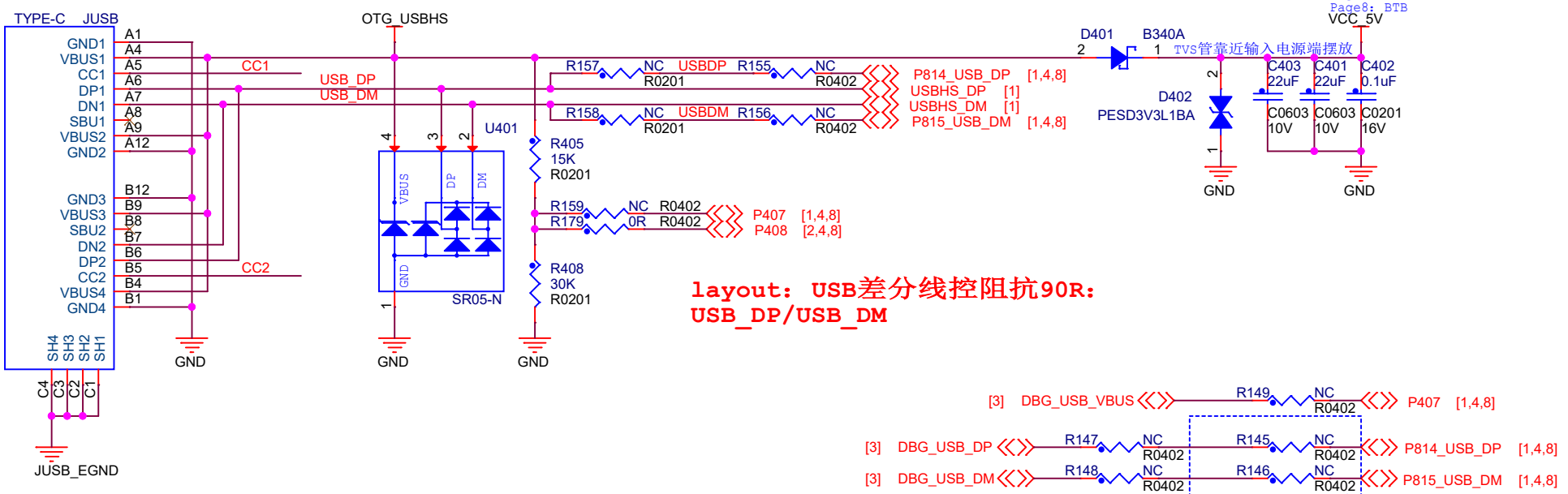
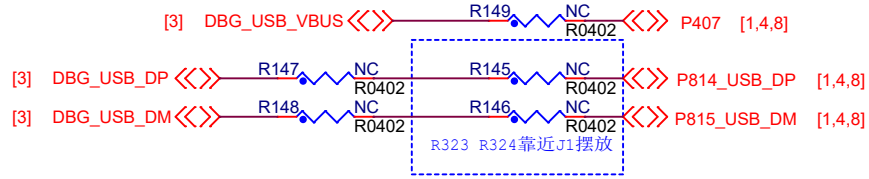
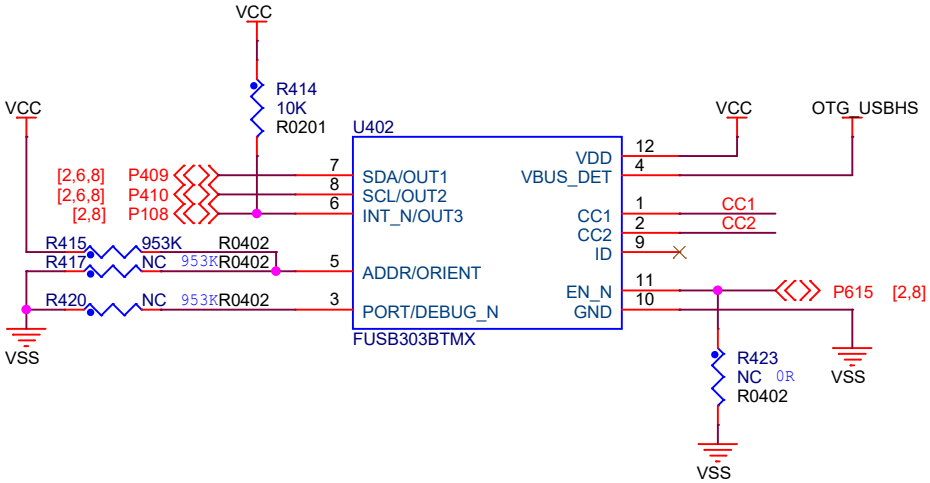
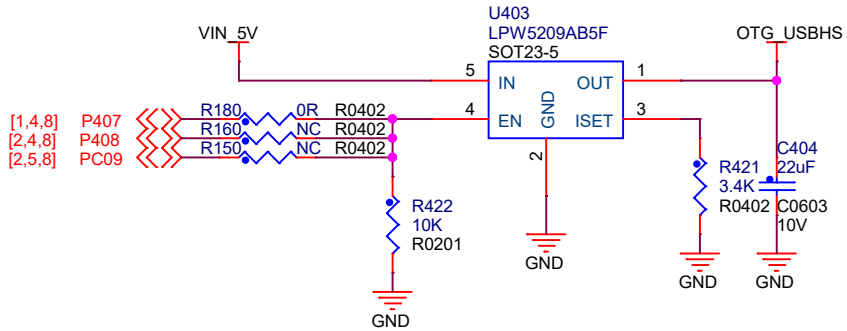


layout: USB差分线控阻抗90R:
USB_DP/USB_DM







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Title 野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用		
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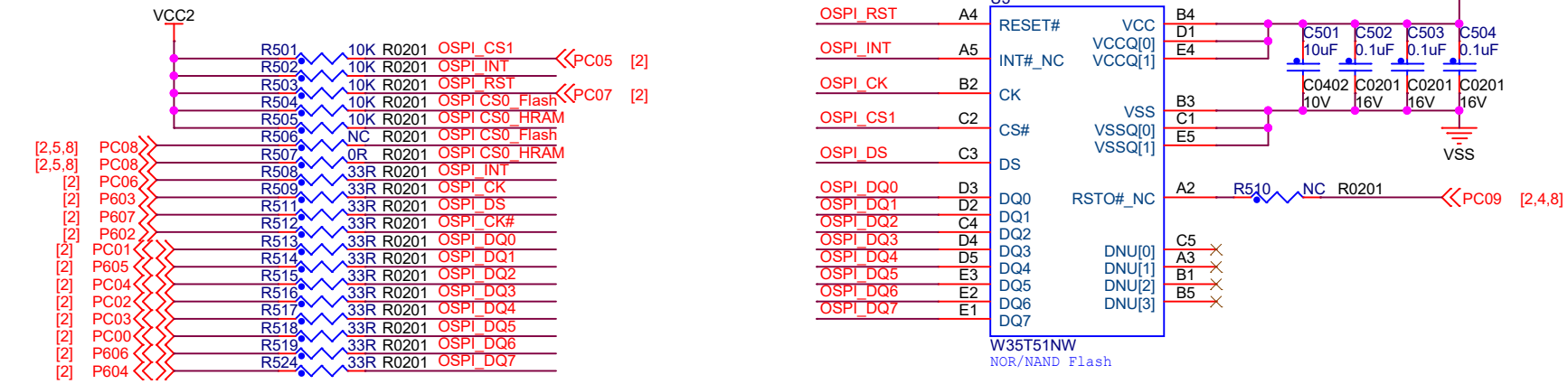
FLASH



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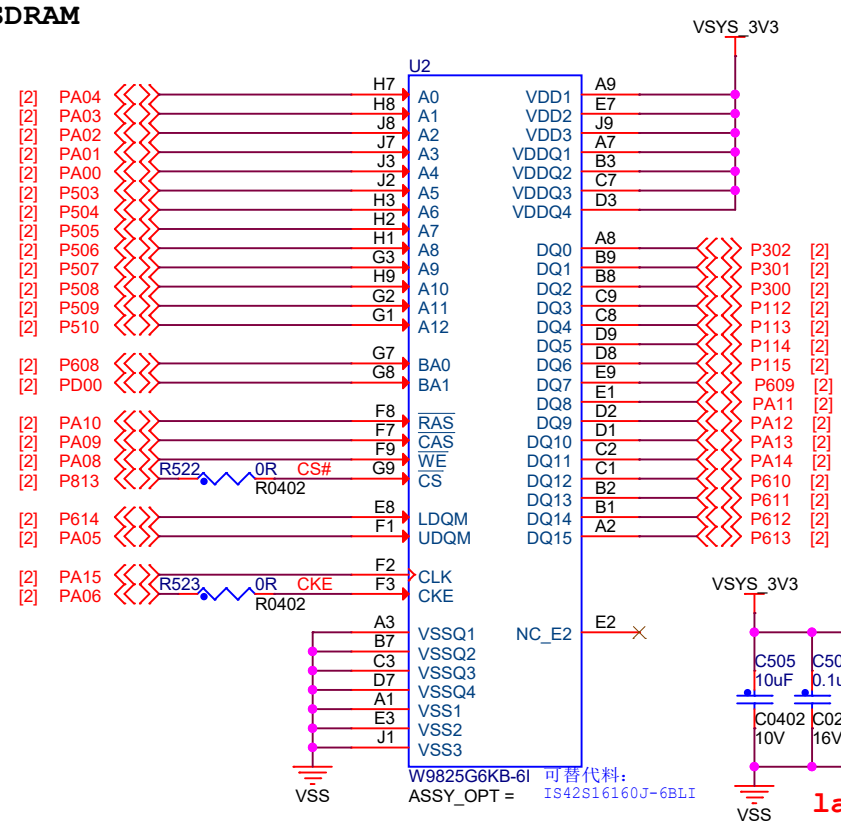
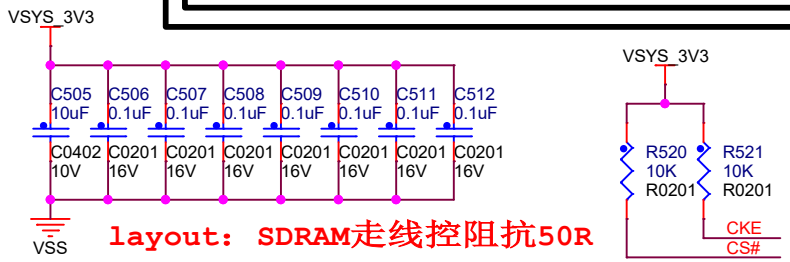
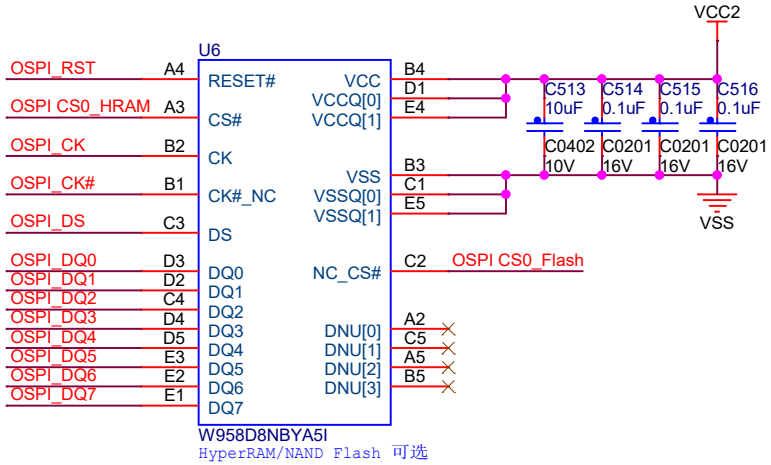
SDRAM





layout: SDRAM走线控阻抗50R



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Title	野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用	
Size A4	Document Number SDRAM	Rev V0.1
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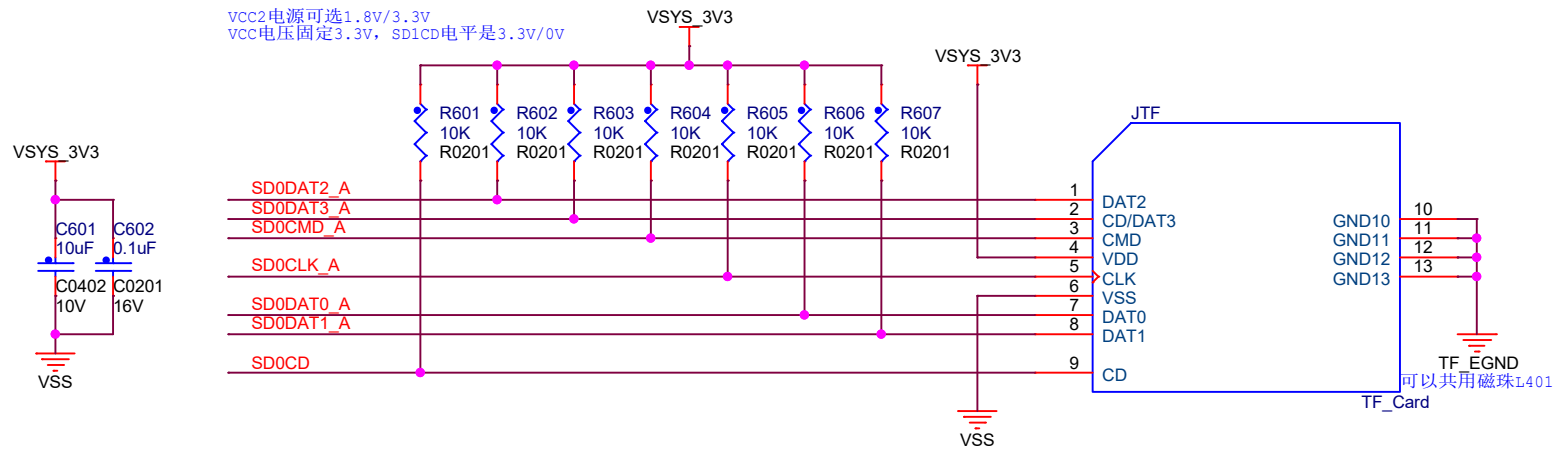
SD卡



Page1: MCU 电源
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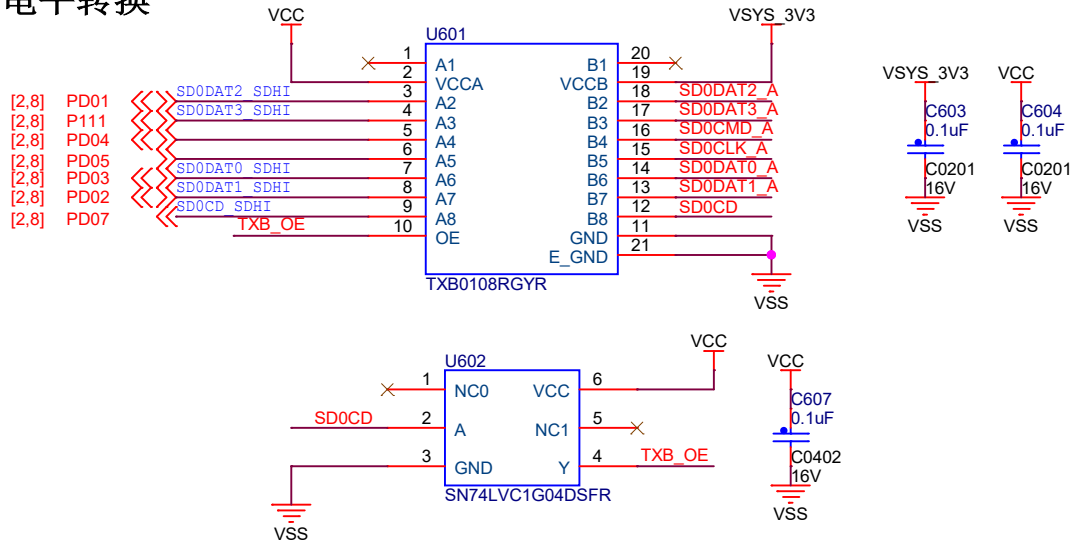
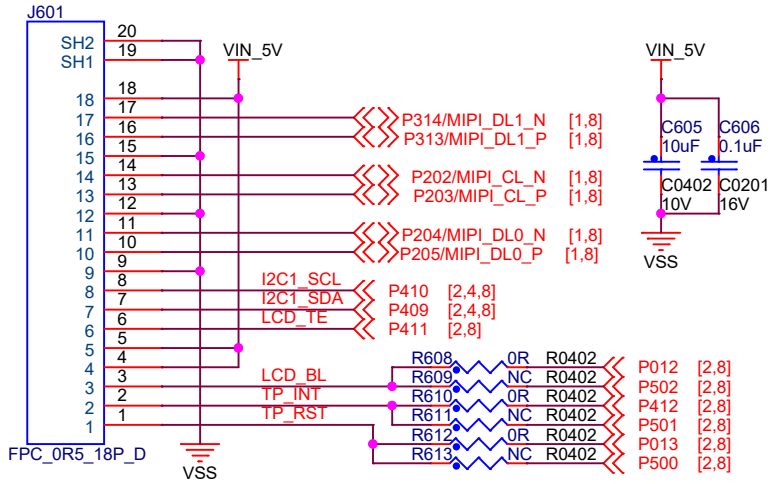
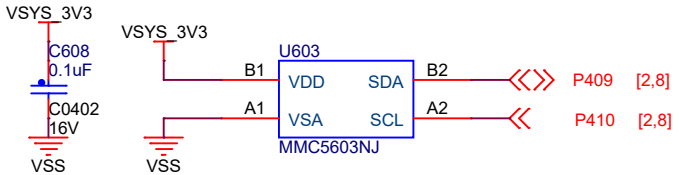
SD卡电平转换



MIPI屏



地磁传感器



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Title
野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用

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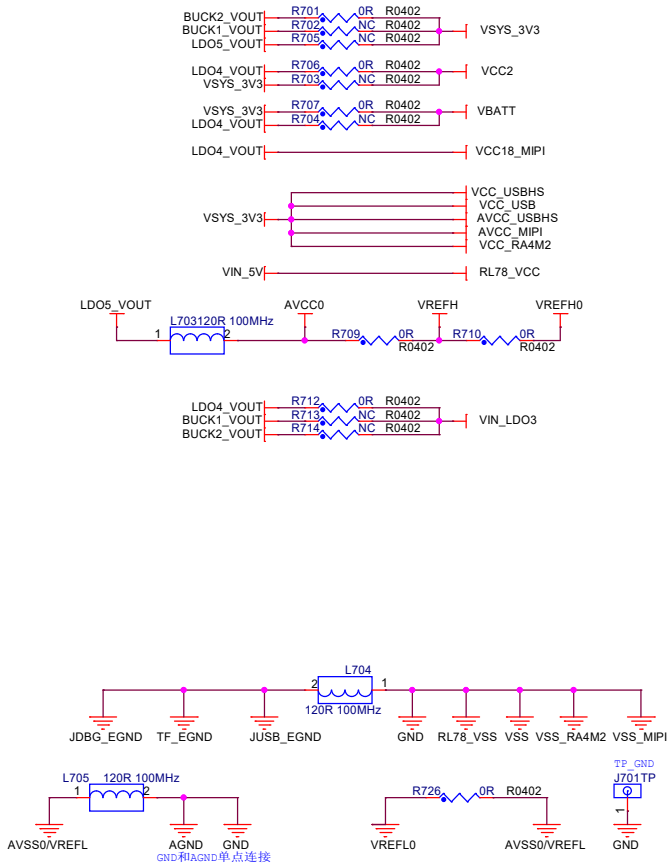
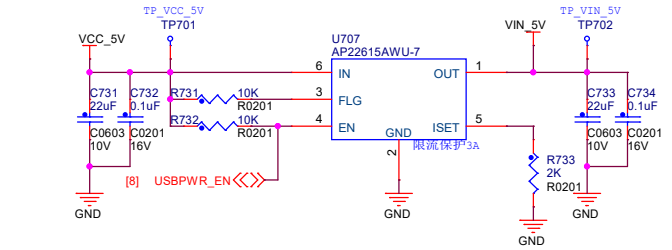
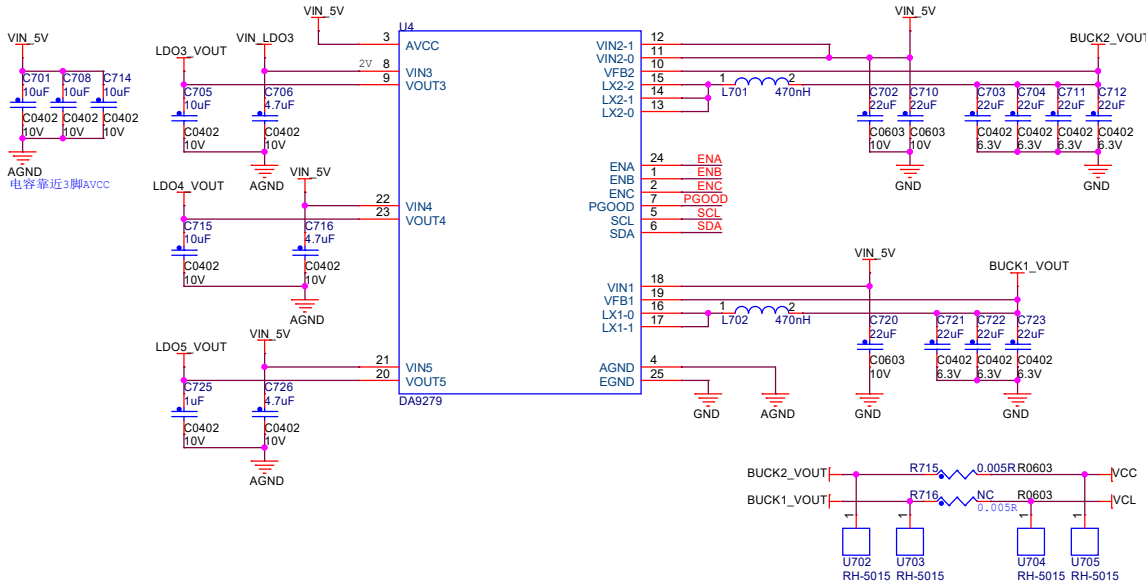
电源分割



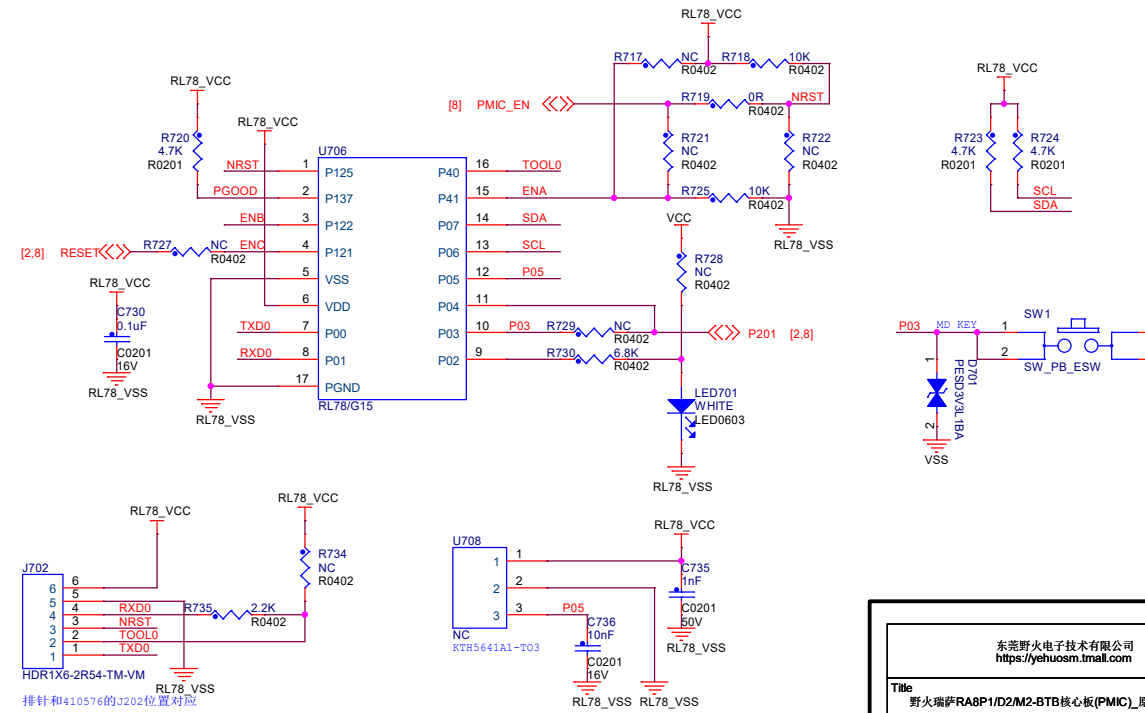
电源保护



PMIC



MCU



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Title 野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用		
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BTB座子

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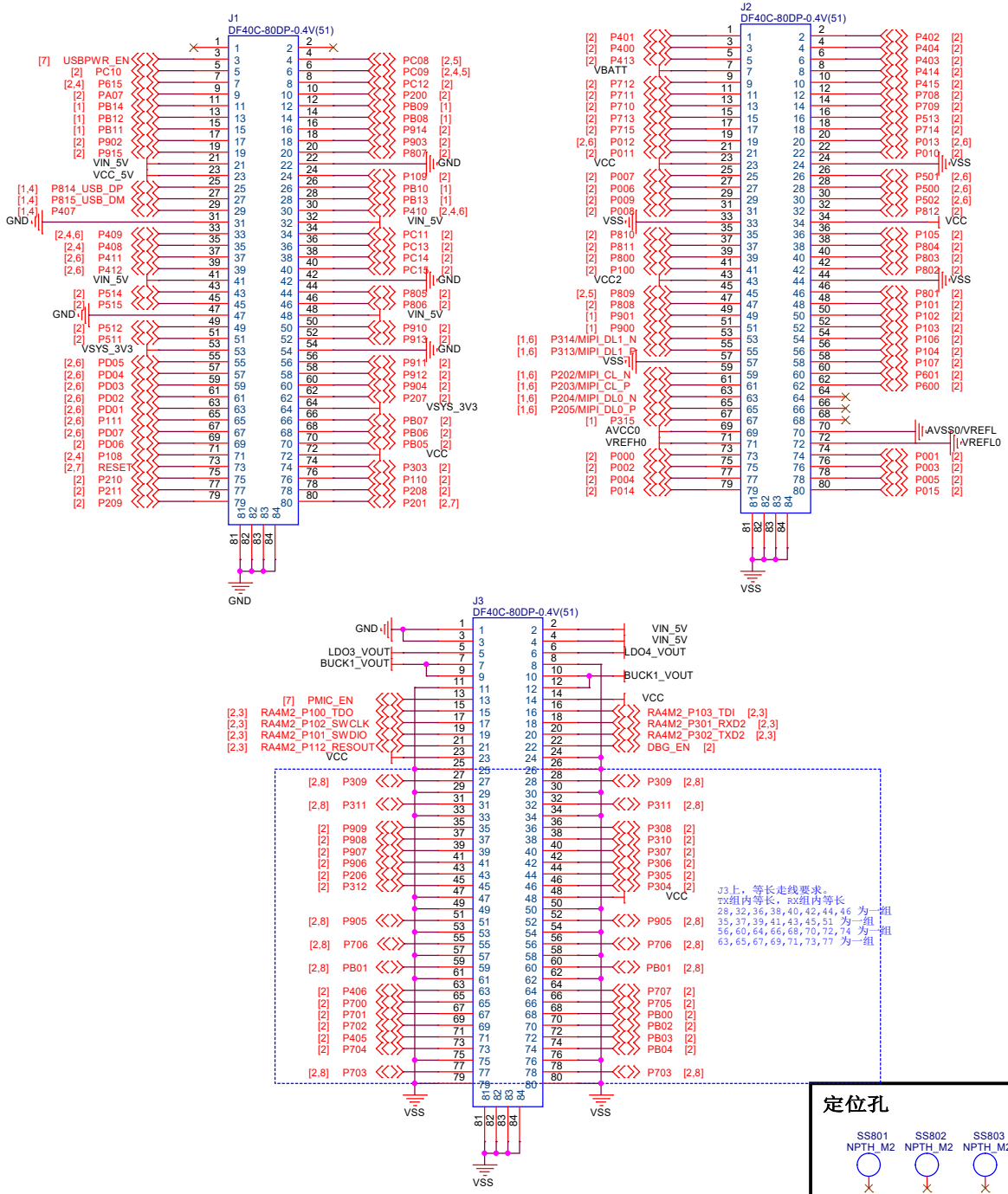

定位孔

SS801	SS802	SS803	SS804
NPTH_M2	NPTH_M2	NPTH_M2	NPTH_M2

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Title: 野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用		
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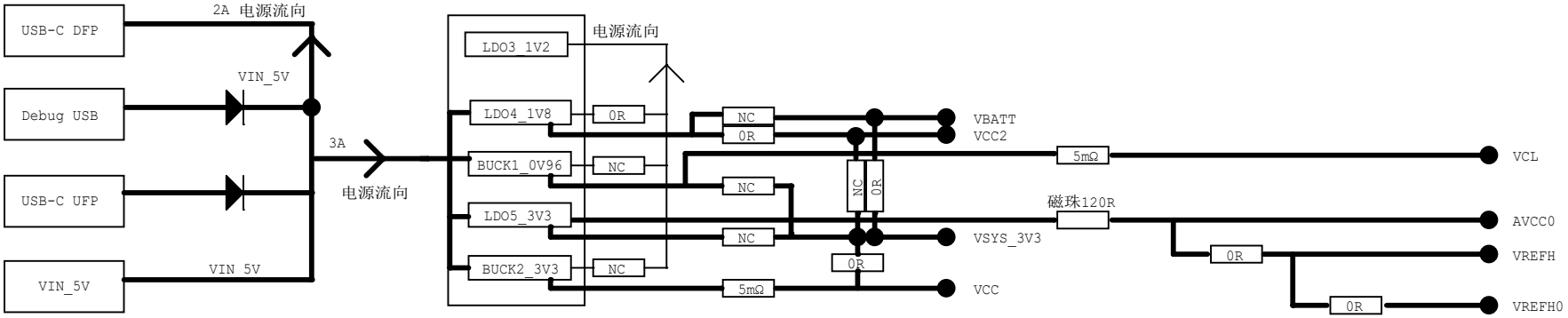
电源拓扑图



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Title 野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用		
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历史版本

版本号	日期	设计	描述
V0.1	2025-05-19	LPC	初始版本

野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用_V0.1

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Title
野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用

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